

Homework 01 - Java Data Types And Variables

Topics Covered

- Data types
- Variables
- Identifiers
- Variable naming rules
- Simple printing using `System.out.println`

Instructions For Students

1. Write all answers in your notebook first.
2. Then write Java programs in your IDE.
3. Use meaningful variable names.
4. Every Java statement should end with semicolon `;`.
5. Submit your `.java` files and notebook answers.

Part A: Very Short Answer Questions

Answer in one or two lines.

1. What is data?
2. What is a data type?
3. What is a variable?
4. Why do we need variables in Java?
5. What is an identifier?
6. Is every variable name an identifier?
7. Can a variable name start with a number?
8. Can a variable name contain spaces?
9. Which data type is used for whole numbers?
10. Which data type is used for decimal numbers?
11. Which data type is used for true/false values?
12. Which data type is used for a single character?
13. Which data type is used for text?
14. What is the difference between `char` and `String`?
15. What is the difference between `=` and `==`? Write only basic meaning.

Part B: Identify The Data Type

Write the correct Java data type for each value.

Value	Correct Data Type
25	
99.50	
'A'	
"A"	
true	
"true"	
50000	
88.75	
'9'	
"9876543210"	

Part C: Valid Or Invalid Variable Names

Write `Valid` or `Invalid`. If invalid, write the reason.

- `int age = 20;`
- `int 1age = 20;`
- `String studentName = "Rahul";`
- `double product price = 99.50;`
- `char grade = 'A';`
- `boolean isPassed = true;`
- `int class = 10;`
- `String city_name = "Delhi";`
- `int totalMarks = 450;`
- `double @salary = 50000.50;`

Part D: Find The Error

Find and correct the mistake.

Question 1

```
int age = "20";
```

Question 2

```
String name = 'Rahul';
```

Question 3

```
char grade = "A";
```

Question 4

```
boolean isPassed = "true";
```

Question 5

```
double price = 99.50
```

Question 6

```
int student age = 21;
```

Question 7

```
int marks;  
System.out.println(marks);
```

Part E: Output Questions

Write the output.

Question 1

```
public class OutputOne {  
    public static void main(String[] args) {  
        int age = 20;  
        System.out.println(age);  
    }  
}
```

Question 2

```
public class OutputTwo {  
    public static void main(String[] args) {  
        String name = "Amit";  
        int age = 22;  
  
        System.out.println("Name: " + name);  
        System.out.println("Age: " + age);  
    }  
}
```

Question 3

```
public class OutputThree {  
    public static void main(String[] args) {  
        int number = 10;  
        number = 20;  
        number = 30;  
  
        System.out.println(number);  
    }  
}
```

Question 4

```
public class OutputFour {  
    public static void main(String[] args) {  
        char grade = 'A';  
        String section = "A";  
  
        System.out.println(grade);  
        System.out.println(section);  
    }  
}
```

Question 5

```
public class OutputFive {  
    public static void main(String[] args) {  
        int maths = 80;  
        int science = 90;  
        int english = 85;  
  
        int total = maths + science + english;  
  
        System.out.println("Total Marks: " + total);  
    }  
}
```

Part F: Coding Practice

Write Java programs for the following.

Program 1: Personal Details

Create variables for:

- name
- age
- city
- course

Print all details.

Program 2: Student Details

Create variables for:

- student name
- roll number
- age
- marks
- grade
- pass status

Print all details.

Program 3: Product Details

Create variables for:

- product name
- price
- quantity
- availability status

Print all details.

Program 4: Employee Details

Create variables for:

- employee name
- employee id
- salary
- department
- permanent employee status

Print all details.

Program 5: Mobile Details

Create variables for:

- brand
- model
- price
- RAM
- 5G support status

Print all details.

Part G: Mini Project

Create a Java program named `StudentProfile` .

Store and print:

- student name
- roll number

- college name
- course name
- semester
- Java marks
- database marks
- web technology marks
- total marks
- percentage
- grade
- pass status

Hint:

```
int total = javaMarks + databaseMarks + webTechnologyMarks;  
double percentage = total / 3.0;
```

Part H: Challenge Questions

These are optional but recommended.

1. Why should mobile number usually be stored as `String` ?
2. Why is `"99.50"` not a `double` ?
3. Why is `"true"` not a `boolean` ?
4. What happens when we assign a new value to the same variable?
5. Write five meaningful variable names using camelCase.

Submission Checklist

Before submitting, check:

- All programs have class names.
- All programs have `main` method.
- All variables have correct data types.
- Variable names are meaningful.
- Code has no missing semicolon.
- Output is properly formatted.